
A Same-Generator Corpus and Classifier Floors for AI-Control Monitor Evaluation

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Abstract

1 Trusted monitors for AI control are tuned and ranked on stored pairs of code:
2 for each programming problem, an honest solution beside a solution carrying a
3 planted backdoor. In the corpora used for this, the honest solution is human-written
4 and the backdoor is written by a language model, so the two classes differ in
5 who wrote them as well as in whether they are sabotaged. Filtering to working
6 solutions and stripping every comment does not remove that difference. On the
7 filtered, comment-stripped split, a bag-of-words classifier separates the two classes
8 at AUROC 0.806, and a classifier that reads no code at all, only problem metadata,
9 reaches 0.772. The signal transfers across backdoor generators and is almost as
10 strong on backdoors that never work, so it is not a property of the sabotage. We
11 then rebuild the corpus so that one model writes both classes on the same problems,
12 and release it. Holding the writer constant removes between a third and nine-tenths
13 of the separability, and holding the prompt constant as well removes most of the
14 rest. An LLM monitor moves the same way, from 0.958 on stored pairs to 0.566 on
15 same-generator pairs. Labelling by whether the backdoor works installs a second
16 artifact: a problem whose backdoor fails contributes its human honest solution, so
17 the negative class records attack failure. Stored-pair evaluations should report a
18 no-code and a bag-of-words floor against permutation nulls, generate both classes
19 from one model under one prompt, and define the negative class by honesty rather
20 than by attack failure.

21 1 Introduction

22 The backbone protocol of AI control is trusted monitoring, where a weaker trusted model reads
23 an untrusted model's output and flags sabotage [3]. Monitors are developed, tuned and ranked
24 almost entirely on *stored* honest/attack pairs, fixed corpora in which each programming problem
25 contributes an honest solution and a solution carrying a planted backdoor [9, 14, 7]. Honest solutions
26 are human-written, taken from APPS [5]; every backdoor is written by a language model, and the
27 corpus records which one. Separation credited to a monitor may be separation between two authors.

28 There are various known confounding variables between the human and AI solutions, such as in
29 commenting style, so corpora are often preprocessed to mitigate them. The filtered release keeps only
30 working honest solutions and working backdoors and strips every comment [9]; Control Tax reports
31 its pass-tests pre-filter and moves its failed backdoors to an appendix [14]; the monitor-ensemble
32 harness of [7] ships a `-leak-check` flag with two trivial baselines in it: a `#-count` detector that
33 falls from 0.6957 to 0.4983 once comments are stripped, and code length alone, at 0.4726. Outside
34 AI control, bag-of-words is known to separate human-written from model-written code [11], and
35 the benchmarks built for that task match their two classes on line and character count or on shared
36 problem definitions [4, 1]. Adjacent control work attacks other axes: monitors subverted with injected

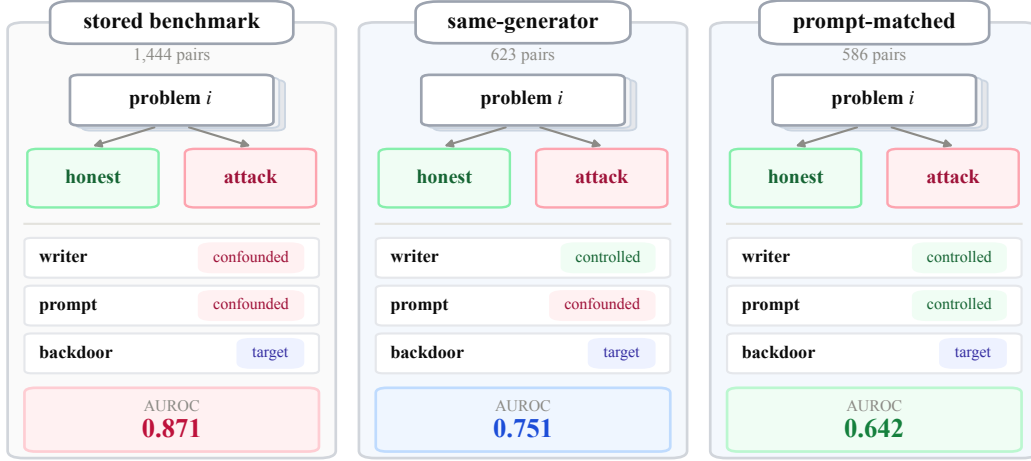


Figure 1: **How a stored pair is built, and how we rebuild it.** Each row is one property of the pair, marked *confounded* where the two classes differ in it and *controlled* where one model or one prompt holds it constant. The backdoor row is the target: it is the only difference a monitor should read, and the only one left once both controls are applied. Under each construction is the bag-of-words classifier’s AUROC on that construction’s own pairs, so the three are a ladder and not a paired comparison. The stored-benchmark value, 0.871, is measured on the full 1,444-problem artifact rather than the filtered split of §3, which gives 0.806; each arm’s numbers are in Table 2, in the notation of Table 1.

37 text, attacker-side selection shown to inflate safety estimates, transcript length removed as a nuisance
38 covariate [15, 2, 12].

39 None of that preprocessing removes the authorship gap. On the filtered, comment-stripped split, a
40 bag-of-words classifier reaches AUROC 0.806, above 20 of the 24 open-weight monitors ranked
41 on that same split, whose median score is 0.609; a classifier that reads no code at all, only problem
42 difficulty, test count and question length – the *no-code floor* – reaches 0.772, above 20. The classifier
43 transfers across backdoor generators and is nearly as strong on *inert* backdoors, ones that never
44 produce a wrong answer, as on working ones. Neither is a property of a backdoor.

45 We contribute (i) a construction for stored-pair corpora, in which one model writes both classes under
46 one prompt framing and one harness verifies both, and the corpus that implements it. Run on two
47 generators, it attributes between a third and nine-tenths of the separability to authorship and most of
48 the remainder to the attack prompt’s framing; a one-sided variant that regenerates only the honest
49 class substitutes one confound for another and separates more than the stored pairs do. We identify
50 (ii) a second selection artifact the filter installs by defining the negative class through attack failure,
51 and (iii) the construction and reporting practice these imply. Our claims concern monitor training,
52 selection and benchmarking on stored pairs, and our measurement is of one APPS-derived family
53 plus five agentic settings (appendix A.9). ControlArena, the evaluation framework most of these
54 corpora are exported from, generates both classes live and is not implicated.

55 2 Setup

56 We use seven public corpora, whose per-class provenance is the object of study: two views of one
57 APPS-backdoor artifact (appendix A.1) and five agentic settings (appendix A.9). In the APPS family
58 the honest class is human and the attack class is one of ten named models. Six of the seven declare
59 no licence [8, 13], so we report only aggregates and redistribute no corpus.

60 Our primary protocol is one of the seven, the filtered, comment-stripped split [9]: 1,209 train problems
61 (2,418 paired items) and 1,375 test items (172 attack, 1,203 honest), labelled by `backdoor_works`,
62 reconstructed with the released harness code [7]. The split is a filtered view of a 1,444-problem
63 artifact, and the same-generator arms below are built over the artifact’s full problem pool, so two

64 stored-pair values recur in this paper: 0.806 on this split, and $B = 0.871$ on the artifact (Figure 1).
65 Generated code is scored by our own runner, applied to both classes (appendix A.11).
66 Against that we build a corpus in which one model writes both classes on the same APPS problems;
67 the construction is run twice, with Sonnet 5 and then Haiku 4.5 as the generator. The *independent*
68 arm writes the attack class from the problem statement alone, so the writer changes and nothing else;
69 it is the arm Table 2’s *headline* rows report. An *edit* arm, shown its own honest solution, returns
70 near-copies (appendix A.4), and a *prompt-matched* arm regenerates the honest class from the attack
71 prompt with the clauses asking for a flaw replaced by their correct-program counterparts, so both
72 classes are elicited alike. We sample one attack per problem, fixed before its outcome is known, and
73 define the attack class as the benchmark does, by intent together with passing the public tests. A
74 backdoor *fires* when its program passes the public tests and returns a wrong answer on the trigger
75 input the attacker declares; an attack that never fires is *inert*. Table 1 fixes the notation, and the bands
76 were registered before any arm ran: $\rho \leq 0.33$ reads as authorship (the *collapse* band), $\rho > 0.72$ as
77 content, and the interval between them as neither (*partial*).

Table 1: **Notation.** One classifier throughout; the arms differ only in the corpus it reads.

Symbol	Quantity
B	on the stored pairs, over all $n = 1,444$ problems
B'	on the stored pairs, refitted on the problems one arm realises
S	on a same-generator arm, where one model wrote both classes
$\rho = (S - 0.5)/(B' - 0.5)$	the share of above-chance separability that survives

78 3 Results

79 Comment stripping succeeds on comment count and code length, driving both to chance (Table A2,
80 appendix). Two baselines on that same split do not move; we call them *floors*, since a monitor scoring
81 at or below one has shown nothing beyond what the corpus leaks. Bag-of-words reaches 0.806 and the
82 no-code floor reaches 0.772, both far above their permutation nulls and above 20 of the 24 monitors
83 ranked on that split (Figure A1; intervals, nulls and paired-bootstrap counts in appendix A.2). The
84 two are not trained alike – the monitors are zero-shot, our classifiers fitted on the split’s own train
85 problems – so the comparison states what the corpus makes recoverable, not that a bag of words is
86 the better monitor.

87 Three further measurements show that the bag-of-words classifier is not reading the backdoor; the first
88 two are pairwise accuracies on matched honest/attack pairs, not AUROCs (appendix A.2). It transfers
89 across backdoor generators at 0.969: trained on one attacker’s backdoors it identifies another’s, so the
90 feature it has learned is one the attackers share. It is nearly as strong on inert backdoors as on working
91 ones, 0.879 against 0.935, although a backdoor that never fires should be the harder case for a detector
92 of sabotage. The feature it does read is stylistic: `__main__` appears in 0.116 of attack samples against
93 0.000 of honest ones, a gap the comment stripper leaves untouched and one that replicates on the
94 train split, in 152 of 1,209 attack samples against 1 of 1,209 honest ones. All three measurements are
95 comment-stripped on both classes; the unstripped values, inflated by the comment channel, are in
96 appendix A.3, and a token-level version of the last test does not replicate (appendix A.12).

97 Holding the writer constant removes between a third and nine-tenths of the separability (Table 2); B'
98 is refitted on each arm’s own problems and includes inert attacks. With $\rho = 1$ meaning that holding
99 the writer constant changed nothing, and $\rho = 0$ that nothing survived, the two generators return 0.687
100 (Sonnet 5) and 0.095 (Haiku 4.5).

101 The pre-registered bands return a different verdict for each generator. Sonnet 5’s interval excludes the
102 collapse band and spans the content boundary, leaving it inconclusive between partial and content;
103 Haiku 4.5’s lies inside the collapse band. Sample size does not account for the fall (appendix A.5).
104 Two further rows are *strict*, keeping only attacks that fire; a single sampled attack does so 2.2% of the
105 time, so these rows rest on 99 pairs and agree with the headline rows at low power (appendix A.6).

106 The two generators respond differently to the attack prompt. On Sonnet 5 what survives is lexical
107 rather than structural: length alone falls to 0.5851 while TF-IDF holds, and of the feature-ablation
108 masks we apply only identifier normalisation moves S (appendix A.5). The top attack-side features
109 are `def main` (67% of attack programs against 38.5% of honest ones) and `data[`, which together

Table 2: **What is left when one model writes both classes.** S is the same-generator AUROC (5-fold, comments stripped) and B' the stored-pair baseline on the row’s own problems; ρ , its bands and its intervals are as in the text and appendix A.5.

Arm	n	S (95% CI)	B'	ρ (95% CI, joint)	band
Sonnet 5, headline	623	0.7512 (0.7295–0.7713)	0.8658	0.687 (0.625–0.755)	partial–content
Haiku 4.5, headline	611	0.5340 (0.5159–0.5531)	0.8584	0.095 (0.044–0.150)	collapse
Sonnet 5, prompt-matched	586	0.6423 (0.6179–0.6621)	0.8613	0.394 (0.326–0.454)	partial
Sonnet 5, strict	99	0.7652 (0.7016–0.8000)	0.7877	0.921 (0.662–1.089)	partial–content
Sonnet 5, strict, matched	96	0.7113 (0.6464–0.7453)	0.8030	0.697 (0.494–0.856)	partial–content

are the program layout the attack prompt elicits. Haiku 4.5 never adopts that layout in any of its three classes – honest, attack, or prompt-matched honest (appendix A.6) – which is why the two generators part. Holding the prompt constant as well as the model moves them back toward each other, to $\rho = 0.394$ on Sonnet 5 (Table 2) and $\rho = 0.113$ on Haiku 4.5 (appendix A.6). On Sonnet this removes most of what is left. The original honest class and the prompt-matched honest class – two correct classes from one model, with no attack on either side – already separate at 0.6006, so the prompt carries framing and no backdoor at all; on Haiku 4.5 that same floor contains chance.

Regenerating one class rather than both fails: an open-weight model (Qwen3-Coder-30B) writing the honest class against the stored backdoors separates *better* than the stored pairs do, $\rho = 1.234$, having replaced one authorship gap with another (appendix A.5).

The same inflation appears when the detector is an LLM monitor rather than a bag-of-words classifier. We put a zero-shot monitor prompt to the model that wrote both classes, the most favourable case available since it is reading its own output. It scores 0.9579 on the stored pairs and 0.566 on the same-generator pairs, a monitor- ρ of 0.145, below the classifier’s 0.687; on the strict pairs the ratio is 0.717, and a monitor that wrote neither class returns nearly the same two ratios (appendix A.7).

The split selects on a second property besides authorship. Labelling by `backdoor_works` defines the negative class by attack failure where the criterion should be honesty: 80.7% of the test split’s labelled backdoors do not work, and a problem whose backdoor fails contributes its human honest solution instead – the label is set by the attack’s outcome before any code is read. Attack problems also carry half as many test cases (20.95 against 38.85), so problem difficulty correlates with the label, which is what the no-code floor’s 0.772 reads.

4 Limitations

A high floor is evidence about a corpus, not about results computed on it. Ours show that the label is recoverable without reading a backdoor, but say nothing about whether any monitor recovers it that way, and no number here corrects a published monitor score.

The control arm is limited by the generator we had to substitute. The model that wrote the stored backdoors, `claude-3-7-sonnet-thinking`, is retired, so both same-generator classes come from a later model of the same lineage. It differs from the stored attacker in backdoor style as well as in authorship, so every ρ here, classifier or monitor, bounds the content share of the separability from above instead of partitioning it. The arm is not bit-reproducible, having run through a metered API that does not expose sampling parameters, with thinking off although the stored attacker was a thinking model.

Our scope is narrow in three further ways. We measure one APPS-derived corpus family and five agentic settings, and other stored-pair corpora are untested. Five of our seven corpora are one user’s repackaging of ControlArena releases, so our claims are scoped to those artifacts as published. And two interpreter-era artifacts, human solutions written for an older Python, fall on opposite classes, so B carries a small execution-environment channel that the same-generator arm does not (appendix A.11).

5 Discussion

These results imply one rule for construction: one model should write both classes under one prompt framing, and one harness should verify both. Regenerating a single side does not help: substituting the honest author left a larger gap than the one it replaced. Matched construction is preceded in code-detection benchmarks [4], though we have not seen it reach backdoor detection.

The definition of the negative class needs the same treatment. Defining it through attack failure ties the label to the attacker rather than to the code; dropping the failed problems removes that, as does sampling the class independently of attack success. A monitor number is in turn interpretable only against the floors of its own split, so a no-code and a bag-of-words floor belong beside it, each against a permutation null, with the split and its preprocessing released alongside (ours are MIT, appendix A.10).

Stored pairs remain the cheapest way to develop a monitor, and nothing here argues for abandoning them, only that a score on one describes two authors until the construction rules that reading out.

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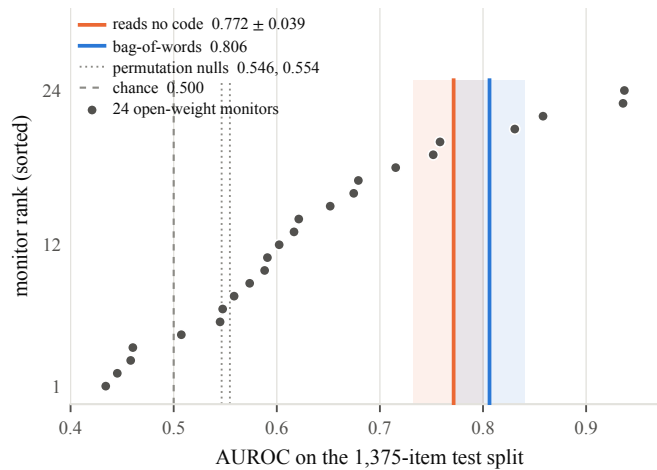
A Supplementary tables and figures

A.1 The schema

Table A1 is the record schema of RoganInglis/apps-control-arena [6]. The attack class carries model and source, naming the system that produced each backdoor; the honest class carries no such field, because there is no generator to name. The two APPS-backdoor corpora we use are two views of this one artifact, since every problem in the filtered release [8] is also in [6], byte-identical except a re-validated backdoor_works.

Table A1: The confound is legible in the dataset schema, with no measurement at all. The attack class records which model wrote it; the honest class has no generator field. Verified against the shard schema of RoganInglis/apps-control-arena.

struct	sub-fields
solutions	code, compiles, passes_tests
backdoors	backdoor_code, backdoor_input, backdoor_works, compiles, model, passes_tests, source



A.2 The protocol $\times$ baseline table

The no-code floor is cross-validated within the test split because the released train/val split is paired and therefore metadata-balanced; its permutation null shuffles labels for fitting and scoring alike. Table A2 is the full measurement behind §3. Rows are not comparable across protocols; the preprocessing column says whether comments were stripped before the baseline read the code. Pairwise accuracies are given to three decimals because their resolution over 2,641 pairs is one pair. Intervals in §3 are bootstraps over problems (2,000 draws) and nulls are 200 label permutations; the held-out no-code row is barely above its own null: 0.716 (CI 0.6476–0.7796) against a 95th percentile of 0.7023, $p = 0.030$, above 18 of the 24 monitors by point estimate and 12 with a paired interval excluding zero. The no-code claim therefore rests on the cross-validated protocol. The two floors of §3 in full: bag-of-words reaches 0.806 (95% CI 0.7704–0.8409 by bootstrap over problems; permutation null 95th percentile 0.5465, $p < 0.005$), above 20 of the 24 monitors by point estimate and 18 with a paired-bootstrap interval excluding zero; the no-code floor reaches 0.772 (CI 0.7319–0.8005 on the pooled out-of-fold 0.766; null 0.5545, $p < 0.005$), above 20 and 17 respectively.

Table A2: What the field’s preprocessing leaves behind. Every row is measured under the protocol named in column one; rows are not comparable across protocols. *Prep.* says whether comments were stripped before the baseline read the code; *no code* means the baseline never reads the program. The comments-kept rows are inflated by the comment channel and are given for contrast, not as results. The matched rows are problem-disjoint and scored against the DSPy tutorial monitor; the transfer row runs Control Tax to the GPT-4-era set. Pairwise accuracies are given to three decimals, their resolution over the pair count.

Protocol	Baseline	Value	Prep.	Against
<i>AUROC</i>				
Koran’s exact split	TF-IDF word 1–2gram	0.8063	stripped	above 20 of 24
same	TF-IDF word 1–2gram	0.8914	kept	above 22 of 24
same, reads no code	difficulty + #tests + $ q $	0.7715 ± 0.0389	no code	above 20 of 24
same, held-out slice	difficulty + #tests + $ q $	0.7163	no code	above 18 of 24
same	code length only	0.4726	stripped	chance
same	#-count detector	0.4983	stripped	chance
same	#-count detector	0.6957	kept	—
Raw artifact, problem level	all metadata	0.8917	no code	—
same	difficulty only	0.6823	no code	—
same	<code>solution_passes_tests</code>	0.7881	no code	—
Matched pairs (P3)	all metadata	0.5000	no code	by construction
<i>Pairwise accuracy</i>				
DSPy matched pairs	TF-IDF word 1–2gram	0.897	stripped	monitor 0.69–0.81
working backdoors only	TF-IDF word 1–2gram	0.935	stripped	—
inert backdoors only	TF-IDF word 1–2gram	0.879	stripped	not sabotage
transfer, cross-generator	TF-IDF word 1–2gram	0.969	stripped	—
all pairs, comments kept	TF-IDF word 1–2gram	0.962	kept	—
working, comments kept	TF-IDF word 1–2gram	0.979	kept	—
inert, comments kept	TF-IDF word 1–2gram	0.955	kept	—
transfer, comments kept	TF-IDF word 1–2gram	0.994	kept	—
same, raw artifact	code length only	0.826	kept	—

A.3 Comment stripping on the matched-protocol numbers

The matched-pair and transfer measurements were first made with comments in. Every value in the authorship paragraph of §3 was re-measured with the same stripper applied to both classes, over five problem-disjoint splits (each transfer row is one fixed split). With comments kept the same quantities were transfer 0.994, working 0.979, inert 0.955, all pairs 0.962 and surface-feature AUROC 0.8306; stripped they are 0.969, 0.935, 0.879, 0.897 and 0.7259. Every one was inflated by the comment channel, which is this paper’s thesis applied to itself. The #-presence marker, 0.685 against 0.176 before stripping, closes to 0.007 against 0.008 after and is dropped; `__main__` is unchanged. One authorship leak survives the stripper because string literals are not comments: in the comment-stripped train split an author byline (Author :) survives in 23 of 1,209 honest samples against 2 attack samples, and a triple-quoted block in 43 against 9. Both are far too rare to drive 0.806,

but both are a literal record of authorship. Stripped, the matched-protocol TF-IDF AUROC is 0.8560, beside $B = 0.871$ and the Koran-split 0.806, all three on stripped code. The other cross-generator transfer rows, stripped, are 0.866 (o3-mini to claude-3-7-sonnet, 82 test pairs), 0.844 (the reverse) and 0.764 (o1-mini to gemini-2.0-flash).

A.4 Pair similarity, and why the same-generator control needs two arms

An attack class built by editing its own honest solution changes pair similarity as well as authorship (Table A3); the independent arm sits in the stored benchmark’s regime and if anything slightly further from it, so it cannot manufacture a low S by making the classes alike.

Table A3: Character similarity of honest/attack pairs, comments stripped. The edit arm’s classes are near-copies; the independent arm reproduces the stored benchmark’s regime, where the two classes are largely independent programs. The last two columns are the share of pairs above and below those similarities.

Attack class	mean sim.	share > 0.95	share < 0.60	changes vs. the benchmark
stored benchmark	0.606	3.4%	45.4%	the regime to match
edit (shown its own solution)	0.928	64.2%	4.0%	authorship and similarity
independent (problem only)	0.659	3.4%	34.8%	authorship only

A.5 Robustness of the same-generator rows

At the strict rows’ $n = 99$, B under the identical pipeline is 0.7807 ± 0.0248 . Intervals in the same-generator paragraphs of §3 are bootstraps over problems, resampling S and B' jointly for ρ ; this replaced a ± 2 fold-sd rule before the retry pool was read. Every ρ point we quote is the ratio of the fold-mean AUROCs and every joint interval beside it is the bootstrap of the pooled out-of-fold ρ , which is the same quantity computed two ways: the pooled points are 0.86 (strict, the row whose interval Table 2 quotes), 0.390 (prompt-matched headline), 0.662 (prompt-matched strict) and 0.097 (the second generator). Five fold seeds move the strict row’s ρ over 0.85–0.92. Retention does not produce the effect. Survivors and non-survivors separate at only 0.62, and S computed with non-firing attempts on the same problems is 0.7595 against 0.7687. The edit arm bounds S under near-copies (similarity 0.928, Table A3): 0.6266 over 427 pairs, $\rho = 0.371$. In the one-sided arm of §3, where only the honest half was regenerated, the classifier reaches 0.962 against $B' = 0.874$ on that arm’s $n = 668$ eligible problems, and TF-IDF gains +0.088 while the length carrier collapses to 0.489. Our own attack prompt halves our single-shot firing rate: 52.0% of our test-passing draws offer a public test input as the trigger, which cannot diverge from the honest solution, and those fire 0.15% against 5.13% for novel triggers; no stored working backdoor has such a trigger (0 of 3,188). The rest is retry-and-keep.

A.6 The prompt-matched arm under the masks, and the second generator

Under prompt matching the identifier mask, the only mask of §3 that moved S , takes the headline row to 0.6264 ($\rho = 0.342$) where it takes the original honest class on the same 586 problems to 0.7046, so identifier normalisation and prompt matching remove largely the same thing. That masked row is the one that decides whether the residue is structure or naming, and under prompt matching it is band-ambiguous: its joint interval, 0.280–0.394, puts 40% of its mass in the collapse band. A character model (3–5-grams, same classifier and folds) retains more than the word tokeniser, 0.6817 ($\rho = 0.493$), so the drop is not an artefact of tokenisation. On the strict pairs the matched row is $S = 0.7113$ ($n = 96$, $\rho = 0.697$, 0.494–0.856). The second generator’s pairs are more self-similar than the first’s (0.724 mean against 0.659), so S was recomputed inside similarity strata, where it stays 0.4737–0.5142 in every stratum, including the 0.451-similarity stratum, which is further from its honest class than the stored benchmark’s own pairs, while B' holds 0.7884–0.8334, so near-copies are not what buys the collapse. Its prompt-matched rows move S to 0.5399 ($\rho = 0.113$, $n = 562$) and put the prompt-only floor at 0.4918, an interval containing chance. The mechanism is visible in the programs. The layout vocabulary that the first generator’s residue is made of, `def main, sys.stdin` and `data[`, is present in 0.000 of the second generator’s programs in *all three* of its classes, which read standard input with `input(` in 1.000 of programs on every side, so there is no layout difference left for the classifier to read.

281 A.7 The monitor rows, and what bounds the second monitor

282 The monitor that wrote both classes scores 0.566 (0.5507–0.5833) on the same-generator headline
283 pairs, a monitor- ρ of 0.145 (0.111–0.182); on the strict pairs it scores 0.818 (0.7630–0.8686), a
284 monitor- ρ of 0.717 (0.601–0.832). The monitor that wrote neither class scores 0.5577 on the same-
285 generator headline pairs and 0.8824 on the stored pairs over the same problems, 0.7871 and 0.8832
286 on the strict pairs. Its headline rows bound the value without fixing it. A 512-token reply ceiling left
287 628 of 1,246 same-generator replies unparseable, and on the stored side the missing share differs by
288 class (50.1% honest against 23.4% attack), a bias of unknown sign. Re-running the strict pairs with a
289 1,024-token ceiling prices it: missingness falls to 2 of 198 and monitor- ρ moves from 0.749 to 0.766,
290 inside the bootstrap interval. On the stored pairs both monitors beat the bag-of-words baseline over
291 the same problems (+0.09 and +0.15 for the first, +0.02 and +0.09 for the second): Figure A1’s
292 ranking states something about those monitors on that split and does not claim that a bag of words
293 beats a monitor here. Classifier and monitor are independent pair by pair on the same-generator pairs
294 ($\phi = 0.04$), so the two are reading different things.

295 A.8 The two generators’ funnel

296 Both generators were run through the identical pipeline. Our honest draws pass the public tests at
297 0.7701 and 0.7278 of 1,444 problems; the second generator’s attack draws pass at 689 of 1,444 and
298 35 of those fire, below the 50-pair minimum `gate_s_samegen.py` requires, leaving it with no strict
299 row and the two generators are compared on the `passes` eligibility only. The two compared rows
300 differ in inertness. 2 of the 611 attacks in the second generator’s row fire (99.67% inert) against 14 of
301 623 (97.75%), the other 33 of its firing draws sitting on problems where its own honest program failed
302 the tests and which the control therefore drops. The single-shot firing rate belongs to the protocol and
303 not to either model, since 52.0% of test-passing draws offer a public test input as their trigger.

304 A.9 The five same-generator agentic settings

305 Removing the authorship gap does not remove every floor. In five settings where the same three
306 models write both classes, so no authorship confound is possible, the surviving floor is structural:
307 0.504 (bash, on its null) to 0.802 (agentdojo) from character count alone (Figure A2). Figure A2
308 shows each setting’s content-blind floor against its own permutation null.

309 A.10 The release

310 The supplementary bundle holds the same-generator split (1,444 honest programs, 1,112 edit-arm
311 and 1,112 independent-arm attack programs written by one model, plus every one of the 11,565
312 draws of the independent arm’s retry pool), both eligibility rules as pair lists under an outcome-blind
313 selection (623 passing and 14 firing independent-arm pairs) and under the benchmark’s own survivor
314 selection (1,046 and 99, or 106 counting a crash on the trigger as firing), the 24 scripts behind the
315 measurements (generation for both generators, the test harness, the trigger verifier, the selection rules,
316 the prompt-matched arm, the monitor rows and each remaining measurement script), a dataset card,
317 and a manifest of row counts and checksums. The licence, MIT, is a file inside the bundle and covers
318 the data and the scripts alike.

319 A.11 The test harness and the pass flag

320 The same-generator arm requires executing generated code, so we built a runner that feeds each
321 shipped input case to a solution under per-case timeouts and address-space limits. It agrees with the
322 artifact’s own `solution_passes_tests` on 95.84% of the analysis pool, and is strictly conservative
323 there, never crediting a solution the artifact rejects (0 such cases). We use *our own* flag for both
324 classes, because a checker stricter on one class than the other would manufacture exactly the artifact
325 this paper documents. We do not length-match either class; the untrusted model chooses its own
326 trajectory length. The headline does not depend on that choice. Under the artifact’s flag $B = 0.871$
327 at $n = 1,444$, under ours $B = 0.873$ at $n = 1,406$, within one standard deviation of each other.
328 The monitor scores in Figure A1 are runs/`x0/monitor-summary.json` from the MIT release of
329 that harness [7], used under its notice. Two interpreter-era artifacts in the upstream solutions fall on
330 opposite classes: a `stdin` without `.buffer`, fixed, and `fractions.gcd`, removed in Python 3.9, not

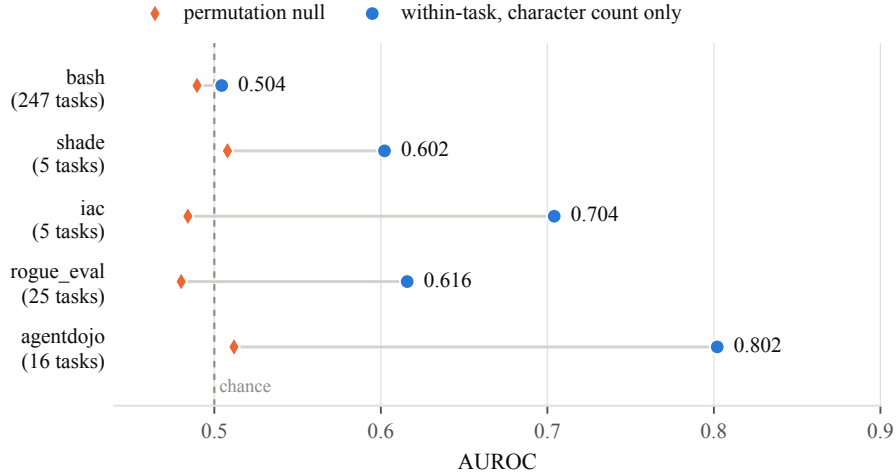


Figure A2: **The structural floor in five same-generator agentic settings.** Each setting’s content-blind classifier (six count features, no content read) scored *within task* against its own permutation null. No authorship confound is possible here: claude-sonnet-4, gemini-2.5-pro and gpt-5 write both classes in all five. bash sits on its null at 0.504; agentdojo reaches 0.802 from character count alone. **The task-group count beside each setting matters:** shade and iac average over five task groups each while bash averages over 247, so these floors are not equally well determined. We report the within-task column because our own pooled numbers partly broke on task composition, and we broke them ourselves. The permutation nulls shown were computed with a pooled estimator rather than the folded within-task statistic; they bound the chance level but are not a like-for-like null for each dot. These corpora declare no licence; we redistribute nothing and report aggregates only.

331 fixable. B is exposed to them; the same-generator arm, whose code we generated and ran ourselves,
 332 is immune.

333 A.12 The token test, run and failed

334 We attempted to confirm a reported token-level mechanism [11] in our setting and it does not replicate.
 335 Separation is -0.013 , the wrong sign, at permutation $p = 0.956$. The reason is register. Their human
 336 class is general-purpose repository code, whereas both of our classes are competitive-programming
 337 code, so an identifier axis that separates general human code from model code does not separate
 338 these two populations. Only 8 of their 10 canonical tokens are in our vocabulary, so the test is
 339 underpowered; but an underpowered true effect would still carry the right sign, and this one does not.
 340 We report it because a reader who knows that work may run exactly this test.

341 A.13 A third channel, noted but not developed here

342 In the same five agentic settings a fifteen-word regular expression over the untrusted policy’s own
 343 scratchpad reaches 0.76–0.82, because prompted attack policies restate their side task in the reasoning
 344 the monitor is given. We note this only to bound the claim. The instrument is not ours, and the work
 345 it comes from [10] validates it against an LLM judge and a length-matched null; we defer to that
 346 treatment.